
Curve-Derived Labels Flatter a Crop Classifier: A Blind Test with Sentinel-2 near Lahore

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https://github.com/mburhankhan101-glitch/crop_project

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Abstract

Crop-type maps of Punjab’s small farms could support food-security and water planning, but there are no public field labels to train or test them on. A common workaround is to label fields from the satellite time series itself, which risks an evaluation that only measures agreement with that labelling. We sampled 100 fields at random in a 10 km square near Lahore, built 5-day Sentinel-2 index series for each, and labelled the winter (Rabi 2026) crop: training fields from clusters of their NDVI curves, and a separate blind test set from true-colour images only. Five classifiers on two feature sets were compared with leave-one-strip-out spatial cross-validation, and the chosen model, logistic regression on 16 hand-made features, was evaluated once on the blind test set. Spatial cross-validation gave 99% accuracy; the blind test gave 89% (24/27, ± 12 points) against a 63% majority baseline. When training labels are derived from the same curves the model learns from, cross-validation overstates accuracy, and an independently labelled test set is needed to measure by how much.

1 Introduction

Pakistani Punjab farms on a rice–wheat rotation: rice in the summer (Kharif) season, wheat and other crops in the winter (Rabi) season. Knowing which crop grows where, field by field, is the starting point for estimating harvests, planning irrigation water and tracking how land use changes. Free Sentinel-2 imagery [2], with a new image every five days at 10 m resolution, makes such maps possible in principle.

Two things make them hard to build and harder to trust. First, the fields are small: in our sample the median field is about one acre, only a few dozen 10 m pixels. Second, there are no public field-level crop labels for the region, so labels must be made by hand, usually by looking at the same imagery the model will learn from. A model trained and cross-validated on labels drawn from its own input curves can score very well while learning little beyond the labelling rule.

This report measures that effect on a small, carefully controlled dataset. Our claim is that a simple model on hand-made time-series features separates the main Rabi classes well, but that cross-validation on curve-derived labels overstates its accuracy, and only an independently labelled blind test shows by how much. The contributions are:

- a labelled set of 100 randomly sampled fields with field outlines, label provenance and a blind test protocol;
- a comparison of five classifiers and two feature sets under spatial cross-validation, with model selection fixed before the test set was opened;
- a measurement of the gap between curve-label cross-validation and the blind test, with an analysis of every test error.

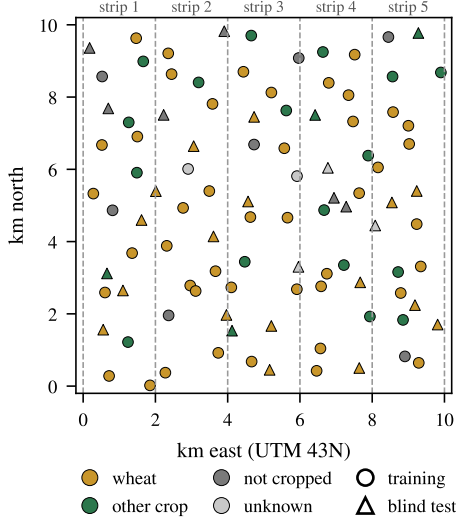


Figure 1: The 100 sample points by Rabi 2026 class. Circles: training; triangles: blind test, six per strip. Dashed lines: the five 2 km strips used as cross-validation blocks.

Table 1: Rabi 2026 labels and the three classes they were merged into; unknown fields were left out.

Class used	Original label	Train	Test
wheat	wheat	44	17
other_crop	other_winter	9	3
	short_winter	6	0
	sugarcane	1	1
	orchard	1	0
not_cropped	non_crop	6	5
	fallow	1	1
(left out)	unknown	2	3

2 Related work

We use Sentinel-2 surface reflectance [2] and standard vegetation indices, NDVI above all [9]. Our training labels follow the idea of Wang et al. [10], who mapped crop types without field-level labels by clustering satellite time series and naming the clusters. They validated against an independent official crop map; we have no such map, so we created a blind test set labelled from a different source (true-colour imagery) and treat the gap between the two evaluations as the main result. Neighbouring fields share soil, water, sowing dates and weather, so randomly split cross-validation tends to be optimistic for spatial data; block cross-validation that holds out whole areas is the recommended alternative [8], and ignoring spatial structure can turn a seemingly useful map model into one with almost no predictive power [7]. We use spatial blocks throughout.

3 Data

Study area and imagery. The study area is a 10×10 km square centred at 31.193 N, 74.303 E, near Raiwind, south-west of Lahore, inside Sentinel-2 tile 43RDQ (Figure 1). We used Level-2A surface reflectance from Microsoft Planetary Computer [5], reading only the image blocks that cover the fields, from 2025-09-08 to 2026-09-25 (67 dates with at least one usable field observation).

Pre-processing. Digital numbers were converted to reflectance as $(DN - 1000)/10000$, the offset used since processing baseline 04.00. Clouds and shadows were taken from the Scene Classification Layer and grown by 2 pixels (20 m), and a field was kept on a date only if at least 80% of its pixels were clear. Field outlines were shrunk by 5 m so that only pixels inside the field were used, and the 20 m bands were resampled to 10 m. Two further checks flag observations the cloud mask misses: a scene-level haze flag (the median blue minus red reflectance of dense fields, $NDVI > 0.5$, above 0.006) and a dip flag (a value more than 0.1 below both neighbouring observations, each within 20 days). Of 5932 field observations, 543 were flagged and left out of the series.

Indices and series. For every field and date we took the median over the field’s pixels of five indices: NDVI (greenness), NDWI (open water), NDMI (moisture, from short-wave infrared), EVI (greenness that saturates less in dense crops) and NDRE (red-edge chlorophyll). Each field’s observations were linearly interpolated onto a regular 5-day grid starting 1 September 2025; gaps longer than 30 days were left empty rather than bridged. Smoothing was tested by holding out observations and predicting them, and no smoothing gave the lowest error, so none was applied. Two gaps are shared by almost every field: early December to late January (winter fog and smog) and mid-July to

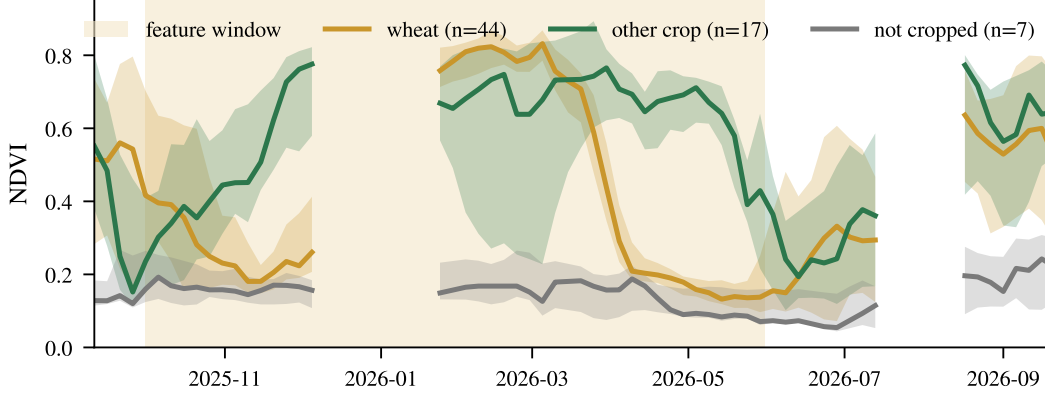


Figure 2: Median NDVI (lines) and interquartile range (bands) of the training fields in each class. The shaded window, October to May, is the period the features are taken from. Wheat and other crops are both green in February; they separate in April, when wheat is harvested. The gaps are the winter fog and monsoon periods.

mid-August (monsoon cloud).

4 Labels

Sampling and outlines. One point was placed at random in each 1 km cell of the square (seed 2026), so that the sample was not biased towards large or easy fields. A field outline was grown from each point over neighbouring pixels whose monthly NDVI through the whole year matched the point's; the similarity threshold (0.08) was calibrated by overlap with three hand-drawn fields. Thirteen outlines that the procedure flagged as doubtful were reviewed against high-resolution imagery: four were kept and nine were labelled `non_crop` (villages, roads, canal banks).

Blind test set. Six points per strip, 30 in all, were set aside before any labelling and labelled from true-colour images only, never from NDVI. Being exact about who labelled what: of the 30, 26 were labelled by an AI model (Claude) from true-colour image chips at the author's request, and 3 of those had had their NDVI profile seen during the outline review, so they are not fully blind. The other 4 were labelled `non_crop` during the outline review, which used both imagery and NDVI.

Training labels. The training fields were clustered by k-means on their NDVI series, with the test fields excluded so their labels could not leak in; the number of clusters (six) was chosen by silhouette score. Each cluster was named from its median curve, and fields that fitted their cluster poorly were read one by one. This gave 65 curve-derived labels; the remaining training fields carry `non_crop` labels from the outline review.

Classes. Several Rabi classes have only one or two training examples, too few to learn from, so they were merged into three classes by what they mean on the ground (Table 1): wheat; other_crop (other winter crops, short winter crops, sugarcane, orchards); and not_cropped (non-crop land and fallow). Fields labelled unknown were left out, which leaves 68 training and 27 test fields. Figure 2 shows why the classes are separable: wheat greens up over December and January and is harvested in late March and April, other crops stay green into May, and the median NDVI of non-cropped land stays below about 0.2 all year.

5 Methods

Features. Set A is the full curve: every index on every 5-day date from 1 October 2025 to 31 May 2026, after dropping dates that are missing for more than half the fields. That leaves 200 features (5 indices $\times$ 40 dates), more than the 68 training fields. Set B has 16 hand-made features: monthly mean NDVI from October to May (8); the Rabi peak NDVI and its date; the number of days with NDVI above 0.5; the minimum NDVI at sowing time; mean NDMI and NDWI in March; peak EVI; and mean NDRE in February. Any remaining missing value is replaced by the median of the training folds only.

Table 2: Balanced accuracy (%) under spatial (leave-one-strip-out) and random 5-fold CV on the 68 training fields; feature counts in brackets. Bold: chosen.

Model	Set A (200)		Set B (16)	
	spatial	random	spatial	random
Majority baseline	33	33	33	33
Logistic reg.	98	97	98	96
SVM (RBF)	98	97	97	94
Random forest	92	95	98	97
Gradient boosting	98	91	98	98

Table 3: Recall of the chosen model per class, spatial CV vs the blind test.

Class	Spatial CV		Blind test	
	<i>n</i>	recall	<i>n</i>	recall
wheat	44	44/44	17	17/17
other_crop	17	16/17	4	3/4
not_cropped	7	7/7	6	4/6
Accuracy		99%		89%
Balanced acc.		98%		81%

Models. All models come from scikit-learn [6]: a majority-class baseline; multinomial logistic regression with an L2 penalty; an SVM with an RBF kernel; a random forest [1] of 500 trees; and gradient boosting [3] with 200 rounds of depth-2 trees, learning rate 0.05. Logistic regression and the SVM use standardised features. To stop the 44 wheat fields from dominating, every model weights classes inversely to their frequency. For logistic regression with weights W and class weights s_y , training minimises

$$\frac{1}{2} \|W\|^2 - C \sum_{i=1}^n s_{y_i} \log p(y_i | x_i; W), \quad (1)$$

where a small C means strong regularisation. C was tuned over $\{0.01, 0.1, 1, 10\}$ for logistic regression and $\{0.1, 1, 10, 100\}$ for the SVM.

Evaluation. The main estimate is leave-one-strip-out cross-validation: each of the five strips is predicted by a model trained on the other four. C is chosen inside each training set by the same leave-one-strip-out scheme (nested cross-validation), so the held-out strip never influences any choice. For comparison only, we also report stratified random 5-fold cross-validation, repeated five times; the difference between the two measures how much neighbouring fields help [8]. Among the eight trained combinations (four models, two feature sets), we chose the simplest one within one standard error of the best spatial score (the one-standard-error rule [4]), preferring fewer features first and then the simpler model. This rule was fixed before the blind test set was opened, and the test set was then used exactly once.

Metrics. We report accuracy, the recall of each class, and balanced accuracy, the mean recall over the $K = 3$ classes,

$$\text{BA} = \frac{1}{K} \sum_{k=1}^K \frac{\#\{i : y_i = k, \hat{y}_i = k\}}{\#\{i : y_i = k\}}, \quad (2)$$

which counts each class equally. For an accuracy p on n fields, the approximate 95% interval is $p \pm 1.96\sqrt{p(1-p)/n}$.

6 Results

Model comparison. Table 2 gives the balanced accuracy of every combination. The best spatial score was 98%, with a standard error of 1.7%, and 7 of the eight trained combinations fell within one standard error of it. The rule therefore chose the simplest: set B with logistic regression ($C = 1$). In spatial cross-validation this model reached 99% accuracy (67/68) and 98% balanced accuracy, against 65% accuracy for always predicting wheat. Random cross-validation gave 96% balanced accuracy, no higher than the spatial estimate.

Blind test. On the 27 blind test fields the model reached 89% accuracy (24/27, ± 12 points, so the true value could be as low as 77%) and 81% balanced accuracy, against 63% (17/27) for always predicting wheat (Table 3, Figure 3). All wheat fields were classified correctly; the three errors were two not_cropped fields predicted as other_crop and one other_crop field predicted as wheat. On the test labels marked high-confidence, the model was correct on 14 of 14.

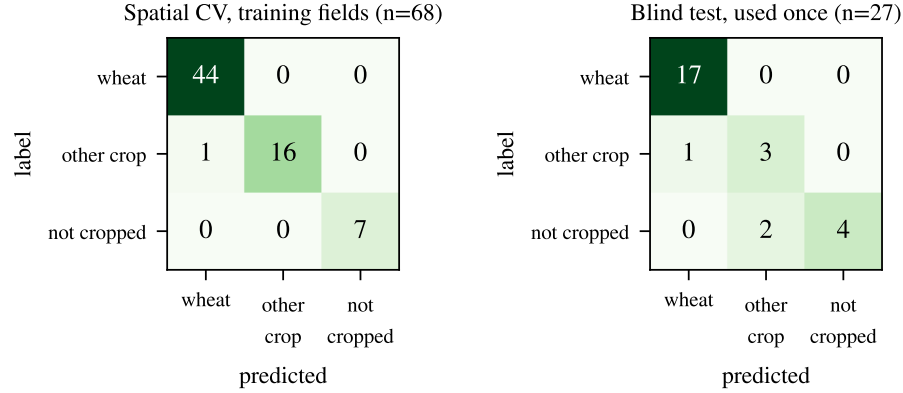


Figure 3: Confusion matrices of the chosen model. Left: spatial cross-validation on the curve-labelled training fields, one error. Right: the blind test, evaluated once, three errors, all in the two smaller classes.

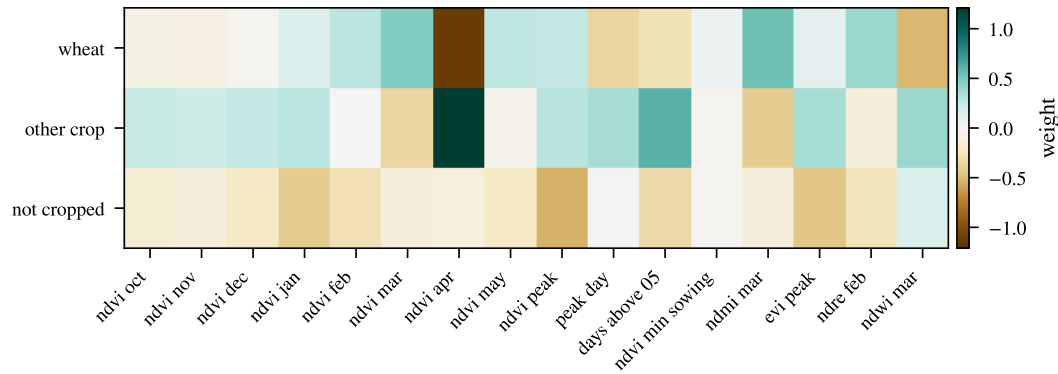


Figure 4: Weights of the final logistic regression on standardised features, one row per class. Green favours the class, brown counts against it. April NDVI dominates: wheat has been harvested by then, other crops have not.

Label confidence. Training only on the 54 high- and medium-confidence labels lowered spatial cross-validation accuracy to 94% (balanced 95%), from 99% (balanced 98%) with all 68 labels.

Feature weights. April NDVI carries the largest weights (Figure 4): strongly negative for wheat and strongly positive for other_crop. Moisture and water in March (NDMI, NDWI) and the number of days above NDVI 0.5 add to the same separation, while low peak NDVI and EVI point to not_cropped.

7 Error analysis

Each blind-test error is shown in Figure 5 against the median curves of its labelled and predicted classes.

g0000: canal-bank trees. The outline covers a line of trees along a canal, labelled non_crop. Its mean NDVI from January to April is 0.64, while no not_cropped training field exceeds 0.26. The training set contains no green non-crop land at all, and the only fields in it that stay green all winter are orchards and sugarcane, labelled other_crop. Given that data, other_crop is the reasonable answer; the error comes from a gap in the training set, not from the model.

g0202: point and outline disagree. The sample point lies on a road verge, which is what the label describes. The outline grown from it, however, covers the green patch beside the verge (mean January–April NDVI 0.56), and the features describe the outline. The model may well be right about the outline; the error lies in the boundary step, where a label for a point was attached to an area it does not describe.

g0600: an early winter crop. This field is green from December, is harvested in early March, about a month before wheat, and then grows a second crop in April and May. The model called it wheat.

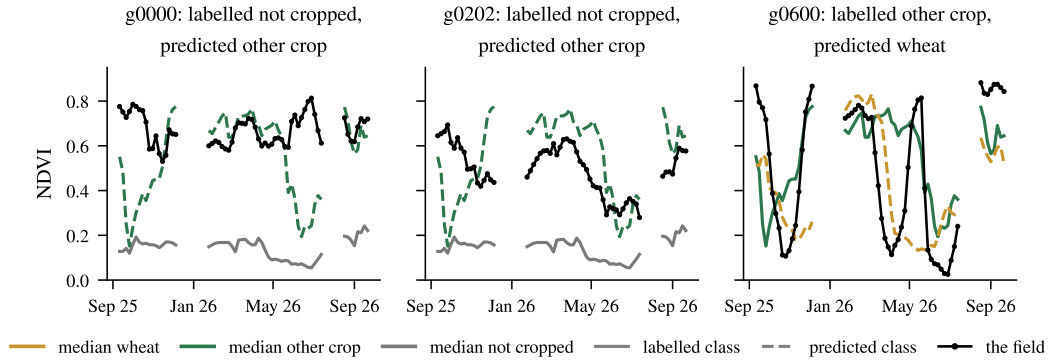


Figure 5: The three blind-test errors (black) with the median training curves of their labelled class (solid) and predicted class (dashed).

The only spatial cross-validation error (g0608, a short winter crop) is the same confusion: a crop that is green through the winter and gone by April looks like wheat to features that mostly ask whether a field is green in winter and bare in April. This is a genuine model weakness; a feature for the timing of the harvest, such as the date NDVI falls below half its peak, would be a natural addition.

8 Discussion and limitations

The gap is the result. Accuracy fell by 10 points from spatial cross-validation to the blind test, and balanced accuracy by 17 points. The training labels were made by clustering the same NDVI curves the models learn from, so cross-validation largely measured how well a model can recover that clustering, which seven of eight combinations did almost perfectly. The blind test labels came from a different source, true-colour images, and so measure agreement with an independent reading of the ground. The two errors on not_cropped fields show the gap concretely: one comes from a kind of land the curve-based training set never contained, the other from a mismatch between a point label and its outline.

Simple was enough. With 200 features for 68 fields, set A gives a model enough freedom to memorise the training set, yet regularised models on it scored no better than those on the 16 hand-made features of set B. A few moments in the season, the January green-up, the March peak and the April harvest, carry nearly all the information, and the hand-made features encode them directly. Because the differences between the top models were smaller than the noise in the estimate, the one-standard-error rule picked the simplest, which is also the easiest to interpret.

No neighbour effect could be measured. Random cross-validation was not more optimistic than spatial cross-validation here. With scores near the ceiling there is little room for a gap, so this result neither confirms nor rules out spatial dependence among these fields.

Limitations. We group them by the kind of validity they threaten. *Internal:* the training labels are circular, as discussed above. *Construct:* the test labels are an AI model's reading of true-colour images rather than field visits, 3 of them were made after the field's NDVI had been seen, 4 come from an outline review that used NDVI, and a point label does not always describe its outline. *External:* one 10 km square and one season, so the results may not hold where crop calendars or field sizes differ. *Statistical:* with 27 test fields, one field is about 4 points, and the 95% interval is ± 12 points; differences of a few fields between models cannot be resolved. Finally, the test set has now been used: any change made in response to these errors must be judged on new blind fields, not on these.

9 Conclusion and future work

A logistic regression on 16 hand-made Sentinel-2 features separated wheat, other crops and non-cropped land near Lahore with 89% accuracy on a blind test set, against 63% for the majority class, while cross-validation on curve-derived labels had suggested 99%. The difference is the price of labelling from the model's own input, and only an independent test set revealed it. Next steps are to

add green non-crop land and early winter crops to the training set, to check labels in the field, to fill the winter and monsoon gaps with Sentinel-1 radar, which sees through cloud, and to test the model in a second district.

Reproducibility

Code, labels and outputs are at https://github.com/mburhankhan101-glitch/crop_project; the results in this report come from commit bf08937. The README lists the commands in order. Sampling used seed 2026 and all models seed 0; the software was Python 3.14.7 and scikit-learn 1.9.1. Every number in the text is generated from the outputs by `s2_paper.py`.

Use of AI tools

The code was written with an AI coding assistant (Claude Code) and run and checked by the author. 26 of the blind test labels were made by Claude from true-colour images at the author's request, 3 of them not fully blind, and the training labels were proposed by Claude from the k-means clusters and reviewed by the author. The text of this report was drafted by Claude at the author's request and reviewed by the author.

Acknowledgements

Contains modified Copernicus Sentinel data (2025–2026), accessed through Microsoft Planetary Computer.

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